

# Mohammad El Prince

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LinkedIn | GitHub | Portfolio

## PROFESSIONAL SUMMARY

### Cloud & Full-Stack Engineer | AWS Serverless

Builder with 3+ years of hands-on software development, including 1.5 years shipping production Amazon Web Services (AWS) systems at Amazon: 20+ serverless applications (Python, TypeScript, Lambda, CDK) and a full-stack quality management platform (React 19, tRPC, DynamoDB) adopted across 8+ North American fulfillment centers. AWS Certified Solutions Architect. Comfortable owning the entire lifecycle solo: requirements, architecture, infrastructure as code, CI/CD, automated testing, security hardening, and production monitoring.

## TECHNICAL SKILLS

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|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages               | Python (incl. pandas), TypeScript, JavaScript, SQL, HTML/CSS                                                                                                                                                                                                                                                                                                                                                          |
| AWS                     | Lambda, API Gateway patterns, DynamoDB (incl. Global Secondary Index design), S3, EventBridge, CloudFront, Lambda@Edge, Simple Email Service (SES), Simple Notification Service (SNS), Systems Manager (SSM) Parameter Store, Identity and Access Management (IAM) least-privilege policies, CloudTrail, CloudWatch, Real User Monitoring (RUM), WAF, Route 53, VPC (subnets, NAT, gateway endpoints), Amazon Bedrock |
| Infrastructure & DevOps | AWS Cloud Development Kit (CDK), Infrastructure as Code (IaC), CI/CD pipelines, gradual/canary deployments, Docker and container-based Lambda, Linux/WSL, Git                                                                                                                                                                                                                                                         |
| Frontend & Backend      | React 19, Next.js, tRPC, Node.js, React Query, Zustand, React Router, Tailwind CSS, Cloud-Scape Design System, Nx monorepos, REST APIs, FastAPI, Flask, Supabase, WCAG 2.1 AA accessibility                                                                                                                                                                                                                           |
| Testing & Quality       | Vitest, property-based testing (fast-check), Pytest, ESLint, snapshot testing, Storybook, React Testing Library, CI-integrated test suites, mypy type checking                                                                                                                                                                                                                                                        |
| AI / ML                 | Retrieval-Augmented Generation (RAG), Amazon Bedrock (LLM integration, Knowledge Bases, S3 Vectors), LangChain, vector embeddings, prompt engineering, streaming LLM responses, AI-assisted development workflows, TensorFlow / Keras neural networks, model training and validation                                                                                                                                  |

## PROFESSIONAL EXPERIENCE

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|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <b>Amazon</b> , Ontario, Canada (Hamilton, St. Thomas, London, Mississauga)                                                                         | 05/2021 – Present |
| <i>Global e-commerce and logistics; roles within the North American Fulfillment Center network (warehouse operations, quality, and automation).</i> |                   |

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|-------------------------------------------------------------------------|-------------------|
| <i>Data Analyst   Cloud &amp; Full-Stack Engineering, AI Automation</i> | 02/2025 – Present |
|-------------------------------------------------------------------------|-------------------|

- Designed, built, and operate 20+ production serverless applications (Python/pandas, TypeScript, AWS Lambda, EventBridge, CDK) that ingest and process operational data from S3 and internal APIs and deliver real-time alerts to managers across North American Fulfillment Centers via Slack, SES, and SNS.
- Architected and shipped a multi-region full-stack quality management platform (React 19, TypeScript, tRPC, DynamoDB, AWS CDK): global CloudFront/Route 53/WAF stack with edge authentication (Lambda@Edge) plus regional API and data stacks, role-based access control, and defect analytics dashboards; launched the MVP at a pilot site and scaled to 8+ locations on user adoption.
- Designed a deterministic scheduling algorithm that auto-generates weekly indirect-role assignments from training and shift data, balancing load across associates while enforcing coverage priorities and weekly hour caps, replacing a fully manual planning process.
- Built an LLM-powered Slack bot on Amazon Bedrock that answers operational questions from S3 data, covering the full integration: IAM permissions, SSM-based secret management, prompt design, and response formatting.
- Hardened production infrastructure with least-privilege IAM policies, VPC gateway endpoints, CloudTrail data-event logging, S3 server access logging, and runtime secret retrieval from SSM Parameter Store (eliminating credentials from CloudFormation templates).
- Enforced 100% line, branch, and function test coverage across all 6 monorepo packages using Vitest, property-based testing (fast-check), snapshot and Storybook tests, and WCAG 2.1 AA accessibility checks, validating every commit through a multi-stage CI/CD pipeline with promotion blockers.
- Owned every project end-to-end as sole developer: stakeholder requirements, architecture, infrastructure-as-code, CI/CD pipelines, gradual deployments, debugging production issues (Lambda cold starts, CloudFront caching, cross-region deployments), and CloudWatch RUM monitoring.

- Modernized a script-based team by introducing serverless architecture, Docker-based development, and IaC practices through architecture reviews and shared design documentation.

Operations Lead / Quality Assurance Specialist

05/2021 – 02/2025

- Led fulfillment teams of 50+ associates; performed root-cause analysis of quality and delivery defects and drove systemic fixes. This frontline context now shapes which problems the tools above solve.

University of Guelph, Guelph, ON Graduate Teaching & Research Assistant

01/2019 – 04/2021

- Designed and validated experimental research setups, correlated experimental against simulation data, co-authored a peer-reviewed publication, and lectured groups of 75-150 students.

KFUPM, Dhahran, Saudi Arabia Graduate Research & Teaching Assistant

09/2016 – 12/2018

- Conducted multiphase fluid flow R&D with Saudi Aramco: experimental design, CFD modeling, and physics-based validation; co-authored 2 peer-reviewed journal papers.

## TECHNICAL PROJECTS

**Noor AI: Conversational RAG Q&A System** | [GitHub](#) | [Live](#)

Production Retrieval-Augmented Generation (RAG) system for Islamic knowledge Q&A over the Quran and major hadith collections. AWS Bedrock (Claude + Cohere multilingual embeddings), Bedrock Knowledge Base with S3 Vectors, LangChain agent with DynamoDB conversation memory, streaming FastAPI backend on Lambda (Function URL + Lambda Web Adapter, no API Gateway), Next.js/Tailwind frontend behind CloudFront, all provisioned as 5 AWS CDK stacks. Citation-safe retrieval design: corpus pre-split per verse/hadith with precomputed metadata so the LLM cites verbatim and cannot fabricate references.

**Quality Management Platform** (*Amazon Confidential*)

Multi-region serverless web application for tracking, researching, and coaching on inventory discrepancies across NA Fulfillment Centers. Nx/TypeScript monorepo (6 packages): React 19 + CloudScape frontend, tRPC Lambda API, DynamoDB with GSI-backed queries, edge authentication via Lambda@Edge, internal authorization (BRASS/Bindle), and CloudWatch RUM monitoring. Features include defect analytics dashboards, coaching workflows, and an algorithmic weekly role-scheduling planner. Live at 8+ sites and expanding.

**YasMade: E-Commerce Platform** | [GitHub](#) | [Demo](#) | [Live](#)

Production e-commerce platform serving live customers. React (Vite) and Supabase full-stack app with secure checkout, admin dashboard, product management, and Zustand global state with optimistic updates; iterating in production against real business feedback.

## EDUCATION & CERTIFICATIONS

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|-------------------------------------------------------------------------------------------------------|---------|
| <b>AWS Certified Solutions Architect - Associate</b> , Amazon Web Services                            | 03/2026 |
| <b>IBM Machine Learning with Python and Deep Learning &amp; Neural Networks with Keras</b> , Coursera | 03/2026 |
| <b>AWS Certified Cloud Practitioner</b> , Amazon Web Services                                         | 07/2025 |
| <b>Full-Stack Web Development Bootcamp</b> , Flatiron School (Python & JavaScript)                    | 10/2023 |
| <b>MEng, Mechanical Engineering</b> , University of Guelph, GPA 91%                                   | 02/2021 |
| <b>MSc, Mechanical Engineering</b> , KFUPM, GPA 3.95/4.00 (thesis-based)                              | 12/2018 |
| <b>BSc, Mechanical Engineering</b> , KFUPM, GPA 3.92/4.00                                             | 05/2016 |

## PUBLICATIONS

Three peer-reviewed publications in fluid mechanics, heat transfer, and thermodynamics, published under the legal name Mohamed Hussein:

- Two-Phase Flow Mass Transfer Analysis of Airlift Pump for Aquaculture Applications. *MDPI* (2021)
- CFD Modeling of Liquid Film Reversal of Two-Phase Flow in Vertical Pipes. *J. of Petroleum Exploration and Production Technology* (2019)
- Effect of Temperature-Dependent Properties on Fin Performance Under Dehumidifying Conditions. *Int. J. of Refrigeration* (2018)